

```
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returninglist <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 521 :opfncid 147 :opresultype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [ 0 0 0 0 0 0 0 0 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile :security_level 0 :num_base_rels 1 :clause_relids (b 1) :required_relids (b 1) :incompatible_relids (b) :outer_relids (b 1) :left_relids (b 1) :right_relids (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause {OEXPR :opno 521 :opfncid 147 :opresultype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 3 :location -1} {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [ 0 0 0 0 0 0 0 0 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relids (b 1) :required_relids (b 1) :incompatible_relids (b) :outer_relids (b 1) :left_relids (b 1) :right_relids (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 2: After pull up At starting of Path Generation:
DETAIL: (QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE true
:isReturn false
:ctelist <>
:table (
{RANGELEFTENTRY
:alias <>
:eref
{ALIAS
:aliasname supplier
:colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "s_acctbal" "s_comment")
}
:rtckind 0
:relid 17308
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals ((
{OEXPR
:opno 1756
:opfncid 1720
:opresultype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{CONST
:consttype 1700
:consttypmod -1
:constcollid 0
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 6 [ 24 0 0 0 0 -128 ]
}
)
)
:location -1
}
)
{ALTERNATIVESUBPLAN
:subplans (
{SUBPLAN
:subLinkType 0
:testexpr <>
:paramids <>
:plan_id 1
:plan_name SubPlan\ 1

```

supplier.s_acctbal > 0

```

:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 0)
:args (
  {VAR
    :varno 1
    :varattno 4
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 4
    :location -1
  }
)
:startup_cost 0
:per_call_cost 24.2625
}
{SUBPLAN
:subLinkType 2
:testexpr
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 4
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 4
    :location -1
  }
  {PARAM
    :paramkind 1
    :paramid 1
    :paramtype 23
    :paramtypmod -1
    :paramcollid 0
    :location -1
  }
)
:location -1
}
:paramids (i 1)
:plan_id 2
:plan_name SubPlan\ 2
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 24.4875
:per_call_cost 0.0025
}
}
))
}
{RANGETBLENTY
:alias
{ALIAS
:aliasname l1
:colnames <>
}
}
:eref
{ALIAS
:aliasname l1
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_lines" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 17349
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
}
{SECURITYQUAIS ((
{ALTERNATIVESUBPLAN
:subplans (
{SUBPLAN
:subLinkType 0
:testexpr <>
:paramids <>
:plan_id 3
:plan_name SubPlan\ 3
:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 2)
:args (
  {VAR
    :varno 2
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 1
    :location -1
  }
)
:startup_cost 0
:per_call_cost 14646
}
}
{SUBPLAN
:subLinkType 2
:testexpr
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0

```

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:args (
  {VAR
    :varno 2
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 1
    :location -1
  }
  {PARAM
    :paramkind 1
    :paramid 3
    :paramtype 23
    :paramtypmod -1
    :paramcollid 0
    :location -1
  }
)
:location -1
}
:paramids (i 3)
:plan_id 4
:plan_name SubPlan\ 4
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 13970.6625
:per_call_cost 0.0025
}
)
}
ALTERNATIVESUBPLAN
{subplans (
  {SUBPLAN
    :subLinkType 0
    :testexpr <>
    :paramids <>
    :plan_id 5
    :plan_name SubPlan\ 5
    :firstColType 2278
    :firstColTypmod -1
    :firstColCollation 0
    :useHashTable false
    :unknownEqFalse true
    :parallel_safe false
    :setParam <>
    :parParam (i 4 5)
    :args (
      {VAR
        :varno 2
        :varattno 2
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 2
        :location -1
      }
      {VAR
        :varno 2
        :varattno 3
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 3
        :location -1
      }
    )
    :startup_cost 0
    :per_call_cost 9472
  }
  {SUBPLAN
    :subLinkType 2
    :testexpr
    {BOOLEXPR
      :boolop and
      :args (
        {OPEXPR
          :opno 96
          :opfuncid 65
          :opresulttype 16
          :opretset false
          :opcollid 0
          :inputcollid 0
          :args (
            {VAR
              :varno 2
              :varattno 2
              :vartype 23
              :vartypmod -1
              :varcollid 0
              :varnullingrels (b)
              :varlevelsup 0
              :varreturningtype 0
              :varnosyn 2
              :varattnosyn 2
              :location -1
            }
            {PARAM
              :paramkind 1
              :paramid 6
              :paramtype 23
              :paramtypmod -1
              :paramcollid 0
              :location -1
            }
          )
        }
        :location -1
      }
      {OPEXPR
        :opno 96
        :opfuncid 65
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
          {VAR
            :varno 2
            :varattno 3
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 2
            :varattnosyn 3
            :location -1
          }
        )
      }
    )
  }
}
)

```

```

(PARAM
:paramkind 1
:paramid 7
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
}
:location -1
}
}
:location -1
}
:paramids (i 6 7)
:plan_id 6
:plan_name SubPlan\ 6
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 8871.94
:per_call_cost 0.005
}
}
)
}
)
}
{RANGETBLENTTRY
:alias <>
:eref
{ALIAS
:aliasname orders
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtekind 0
:relid 17337
:inh false
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromC1 true
:securityQuals <>
}
{RANGETBLENTTRY
:alias <>
:eref
{ALIAS
:aliasname nation
:colnames ("n_nationkey" "n_name" "n_regionkey" "n_comment")
}
:rtekind 0
:relid 17285
:inh false
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromC1 true
:securityQuals <>
}
{RANGETBLENTTRY
:alias <>
:eref
{ALIAS
:aliasname *GROUP*
:colnames ("s_name")
}
:rtekind 9
:groupexprs (
{VAR
:varno 1
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varrattnosyn 2
:location 51
}
)
:lateral false
:inFromC1 false
:securityQuals <>
}
{RANGETBLENTTRY
:alias
{ALIAS
:aliasname l2
:colnames <>
}
}
:eref
{ALIAS
:aliasname l2
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenummer" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 17349
:inh false
:relkind r
:rellockmode 1
:perminfoindex 5
:tablesample <>
:lateral false
:inFromC1 true
:securityQuals (
{ALTERNATIVESUBPLAN
:subplans (
{SUBPLAN
:subLinkType 0
:testexpr <>
:paramids <>
:plan_id 7
:plan_name SubPlan\ 7
:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 8)
:args (
{VAR
:varno 6
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6

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        :varattnosyn 1
        :location -1
    }
}
:startup_cost 0
:per_call_cost 14646
}
{SUBPLAN
:subLinkType 2
:testexpr
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 6
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 1
:location -1
}
{PARAM
:paramkind 1
:paramid 9
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
:paramids (i 9)
:plan_id 8
:plan_name SubPlan\ 8
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 13970.6625
:per_call_cost 0.0025
}
}
}
{ALTERNATIVESUBPLAN
:subplans (
{SUBPLAN
:subLinkType 0
:testexpr <>
:paramids <>
:plan_id 9
:plan_name SubPlan\ 9
:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 10 11)
:args (
{VAR
:varno 6
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 2
:location -1
}
{VAR
:varno 6
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 3
:location -1
}
)
:startup_cost 0
:per_call_cost 9472
}
{SUBPLAN
:subLinkType 2
:testexpr
{BOOLEXP
:boolop and
:args (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 6
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 2
:location -1
}
{PARAM
:paramkind 1
:paramid 12
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16

```

```

:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 6
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 6
    :varattnosyn 3
    :location -1
  }
  {PARAM
    :paramkind 1
    :paramid 13
    :paramtype 23
    :paramtypmod -1
    :paramcollid 0
    :location -1
  }
)
:location -1
}
)
:location -1
}
}
:paramids (i 12 13)
:plan_id 10
:plan_name SubPlan\ 10
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 8871.94
:per_call_cost 0.005
}
)
}
)
}
{RANGETBLENTY
:alias
{ALIAS
:aliasname 13
:colnames <>
}
:eref
{ALIAS
:aliasname 13
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 17349
:inh false
:relkind r
:rellockmode 1
:perminfoindex 6
:tablesample <>
:lateral false
:inFromCl true
:securityQuals ((
{ALTERNATIVESUBPLAN
:subplans (
{SUBPLAN
:subLinkType 0
:testexpr <>
:paramids <>
:plan_id 11
:plan_name SubPlan\ 11
:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 14)
:args (
{VAR
:varno 7
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 1
:location -1
}
)
:startup_cost 0
:per_call_cost 14646
}
{SUBPLAN
:subLinkType 2
:testexpr
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 7
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 1
:location -1
}
{PARAM
:paramkind 1
:paramid 15
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
}
:paramids (i 15)
:plan_id 12
:plan_name SubPlan\ 12
:firstColType 23
:firstColTypmod -1

```

```

:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 13970.6625
:per_call_cost 0.0025
}
}
}
(ALTERNATIVESUBPLAN
:subplans (
{SUBPLAN
:sublinkType 0
:testexpr <>
:paramIds <>
:plan_id 13
:plan_name SubPlan\ 13
:firstColType 2278
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 16 17)
:args (
{VAR
:varno 7
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 2
:location -1
}
{VAR
:varno 7
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 3
:location -1
}
)
:startup_cost 0
:per_call_cost 9472
}
(SUBPLAN
:sublinkType 2
:testexpr
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 7
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 2
:location -1
}
}
{PARAM
:paramkind 1
:paramid 18
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 7
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 3
:location -1
}
}
{PARAM
:paramkind 1
:paramid 19
:paramtype 23
:paramtypmod -1
:paramcollid 0
:location -1
}
)
:location -1
}
)
:location -1
}
)
:paramIds (i 18 19)
:plan_id 14
:plan_name SubPlan\ 14
:firstColType 23
:firstColTypmod -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 8871.94
:per_call_cost 0.005
}
)

```

```

    }
  })
}
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17308
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17349
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10 19 20)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17337
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 10)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17285
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17349
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17349
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
}
)

```

```

:jointree
{FROMEXPR
:fromlist (
{JOINEXPR
:jointype 5
:isNatural false
:larg
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
{RANGETBLREF
:rtindex 2
}
{RANGETBLREF
:rtindex 3
}
{RANGETBLREF
:rtindex 4
}
)
):quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 176
}
{VAR
:varno 2
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 3
:location 188
}
)
:location 186
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 213
}
{VAR
:varno 2
:varattno 1

```

supplier ↔ lineitem
lineitem ↔ orders

supplier.s_suppkey =
lineitem.l_suppkey

orders.o_orderkey =
lineitem.l_orderkey

```

:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 226
}
}
:location 224
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 3
:varattno 3
:vartype 1042
:vartypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 3
:location 252
}
}{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location 268
:constvalue 5 [ 20 0 0 0 70 ]
}
)
:location 266
}
{OPEXPR
:opno 1097
:opfuncid 1009
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 2
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 13
:location 284
}
{VAR
:varno 2
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 12
:location 303
}
}
)
:location 301
}
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 776
}
{VAR
:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location 790
}
}
)
:location 788
}
{OPEXPR
:opno 1054
:opfuncid 1048
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 100
:args (
{VAR
:varno 4
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location 814
}
}{CONST
:consttype 1042
:consttypmod -1
:constcollid 100
:constlen -1
:constbyval false

```

o_orderstatus = 'F'

l1.l_receiptdate >
l1.l_commitdate

supplier.s_nationkey =
nation.n_nationkey

n_name =
'ARGENTINA'

```

:constisnull false
:location 823
:constvalue 13 [ 52 0 0 0 65 82 71 69 78 84 73 78 65 ]
}
):location 821
}
)
}
:rarg
{RANGETBLREF
:rtindex 6
}
:usingClause <>
:join_using_alias <>
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 6
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 1
:location 420
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 436
}
}
:location 434
}
{OEXPR
:opno 518
:opfuncid 144
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 6
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 6
:varattnosyn 3
:location 472
}
{VAR
:varno 2
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 3
:location 488
}
}
:location 485
}
)
:alias <>
:rtindex 0
}
:rarg
{RANGETBLREF
:rtindex 7
}
:usingClause <>
:join_using_alias <>
:quals (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 7
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 1
:location 616
}
{VAR
:varno 2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location 632
}
}
)
:location 630
}
{OEXPR
:opno 518
:opfuncid 144
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 7

```

I2.I_orderkey =

I1.I_orderkey

I2.I_supkey <>

I1.I_supkey

I3.I_orderkey =

I1.I_orderkey

AND I3.I_supkey

<> I1.I_supkey

AND

I3.I_receiptdate >

I3.I_commitdate

```

:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 3
:location 668
}
{VAR
:varno 2
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 3
:location 684
}
}
:location 681
}
{OPEXPR
:opno 1097
:opfuncid 1089
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 7
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 13
:location 719
}
{VAR
:varno 7
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 7
:varattnosyn 12
:location 738
}
}
:location 736
}
)
:alias <>
:rtindex 0
}
)
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 1
:resname s_name
:ressortgroupref 2
:resorigtbl 17308
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggssplit 0
:aggno -1
:aggtransno -1
:location 67
}
:resno 2
:resname numwait
:ressortgroupref 1
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 2
:eqop 1054
:sortop 1058
:reverse_sort false
:nulls_first false
:hashable true
}
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>

```


QUERY PLAN

```
Sort (cost=130254146433.64..130254146433.75 rows=43 width=34) (actual time=1241.938..1241.953 rows=0.00 loops=1)
Output: supplier.s_name, (count(*))
Sort Key: (count(*)) DESC, supplier.s_name
Sort Method: quicksort Memory: 25kB
Buffers: shared hit=2431 read=52525
-> GroupAggregate (cost=130254146431.72..130254146432.47 rows=43 width=34) (actual time=1241.914..1241.928 rows=0.00 loops=1)
Output: supplier.s_name, count(*)
Group Key: supplier.s_name
Buffers: shared hit=2425 read=52525
-> Sort (cost=130254146431.72..130254146431.83 rows=43 width=26) (actual time=1241.912..1241.926 rows=0.00 loops=1)
Output: supplier.s_name
Sort Key: supplier.s_name
Sort Method: quicksort Memory: 25kB
Buffers: shared hit=2425 read=52525
-> Nested Loop Semi Join (cost=0.00..130254146430.55 rows=43 width=26) (actual time=1241.905..1241.918 rows=0.00 loops=1)
Output: supplier.s_name
Join Filter: ((i2.l.suppkey <> i1.l.suppkey) AND (orders.o_orderkey = i2.l_orderkey))
Buffers: shared hit=2425 read=52525
-> Nested Loop (cost=0.00..86836156439.96 rows=58 width=38) (actual time=1241.905..1241.915 rows=0.00 loops=1)
Output: supplier.s_name, i1.l.suppkey, i1.l_orderkey, orders.o_orderkey
Join Filter: ((i3.l.suppkey <> i1.l.suppkey) AND (i3.l_orderkey = i1.l_orderkey))
```